a. What proportion of college students obtain a score greater than 0.8 on the mathematics test?

b. If a student's score on the music test is 0.3, what is the probability that his score on the mathematics test will be greater than 0.8?

c. If a student's score on the mathematics test is 0.3, what is the probability that his score on the music test will be greater than 0.8?

9. Suppose that either of two instruments might be used for making a certain measurement. Instrument 1 yields a measurement whose p.d.f. h_1 is

$$h_1(x) = \begin{cases} 2x & \text{for } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Instrument 2 yields a measurement whose p.d.f. h_2 is

$$h_2(x) = \begin{cases} 3x^2 & \text{for } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Suppose that one of the two instruments is chosen at random and a measurement X is made with it.

a. Determine the marginal p.d.f. of X.

b. If the value of the measurement is $X = 1/4$, what is the probability that instrument 1 was used?

10. In a large collection of coins, the probability X that a head will be obtained when a coin is tossed varies from one coin to another, and the distribution of X in the collection is specified by the following p.d.f.:

$$f_1(x) = \begin{cases} 6x(1-x) & \text{for } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Suppose that a coin is selected at random from the collection and tossed once, and that a head is obtained. Determine the conditional p.d.f. of X for this coin.

11. The definition of the conditional p.d.f. of X given $Y = y$ is arbitrary if $f_2(y) = 0$. The reason that this causes no serious problem is that it is highly unlikely that we will observe Y close to a value y_0 such that $f_2(y_0) = 0$. To be more precise, let $f_2(y_0) = 0$, and let $A_0 = [y_0 - \epsilon, y_0 + \epsilon]$. Also, let y_1 be such that $f_2(y_1) > 0$, and let $A_1 = [y_1 - \epsilon, y_1 + \epsilon]$. Assume that f_2 is continuous at both y_0 and y_1. Show that

$$\lim_{\epsilon \to 0} \frac{\Pr(Y \in A_0)}{\Pr(Y \in A_1)} = 0.$$

That is, the probability that Y is close to y_0 is much smaller than the probability that Y is close to y_1.

12. Let Y be the rate (calls per hour) at which calls arrive at a switchboard. Let X be the number of calls during a two-hour period. Suppose that the marginal p.d.f. of Y is

$$f_2(y) = \begin{cases} e^{-y} & \text{if } y > 0, \\ 0 & \text{otherwise,} \end{cases}$$

and that the conditional p.f. of X given $Y = y$ is

$$g_1(x|y) = \begin{cases} \dfrac{(2y)^x}{x!} e^{-2y} & \text{if } x = 0, 1, \ldots, \\ 0 & \text{otherwise.} \end{cases}$$

a. Find the marginal p.f. of X. (You may use the formula $\int_0^\infty y^k e^{-y}\, dy = k!$.)

b. Find the conditional p.d.f. $g_2(y|0)$ of Y given $X = 0$.

c. Find the conditional p.d.f. $g_2(y|1)$ of Y given $X = 1$.

d. For what values of y is $g_2(y|1) > g_2(y|0)$? Does this agree with the intuition that the more calls you see, the higher you should think the rate is?

13. Start with the joint distribution of treatment group and response in Table 3.6 on page 138. For each treatment group, compute the conditional distribution of response given the treatment group. Do they appear to be very similar or quite different?

3.7 Multivariate Distributions

In this section, we shall extend the results that were developed in Sections 3.4, 3.5, and 3.6 for two random variables X and Y to an arbitrary finite number n of random variables $X_1, \ldots, X_n$. In general, the joint distribution of more than two random variables is called a multivariate distribution. *The theory of statistical inference (the subject of the part of this book beginning with Chapter 7) relies on mathematical models for observable data in which each observation is a random variable. For this reason, multivariate distributions arise naturally in the mathematical models for data. The most commonly used model will be one in which the individual data random variables are conditionally independent given one or two other random variables.*

Joint Distributions

Example 3.7.1

A Clinical Trial. Suppose that m patients with a certain medical condition are given a treatment, and each patient either recovers from the condition or fails to recover. For each $i = 1, \ldots, m$, we can let $X_i = 1$ if patient i recovers and $X_i = 0$ if not. We might also believe that there is a random variable P having a continuous distribution taking values between 0 and 1 such that, if we knew that $P = p$, we would say that the m patients recover or fail to recover independently of each other each with probability p of recovery. We now have named $n = m + 1$ random variables in which we are interested. ◀

The situation described in Example 3.7.1 requires us to construct a joint distribution for n random variables. We shall now provide definitions and examples of the important concepts needed to discuss multivariate distributions.

Definition 3.7.1

Joint Distribution Function/c.d.f. The *joint c.d.f.* of n random variables $X_1, \ldots, X_n$ is the function F whose value at every point $(x_1, \ldots, x_n)$ in n-dimensional space R^n is specified by the relation

$$F(x_1, \ldots, x_n) = \Pr(X_1 \le x_1, X_2 \le x_2, \ldots, X_n \le x_n). \tag{3.7.1}$$

Every multivariate c.d.f. satisfies properties similar to those given earlier for univariate and bivariate c.d.f.'s.

Example 3.7.2

Failure Times. Suppose that a machine has three parts, and part i will fail at time X_i for $i = 1, 2, 3$. The following function might be the joint c.d.f. of X_1, X_2, and X_3:

$$F(x_1, x_2, x_3) = \begin{cases} (1 - e^{-x_1})(1 - e^{-2x_2})(1 - e^{-3x_3}) & \text{for } x_1, x_2, x_3 \ge 0, \\ 0 & \text{otherwise.} \end{cases}$$ ◀

Vector Notation In the study of the joint distribution of n random variables $X_1, \ldots, X_n$, it is often convenient to use the vector notation $\boldsymbol{X} = (X_1, \ldots, X_n)$ and to refer to $\boldsymbol{X}$ as a *random vector*. Instead of speaking of the joint distribution of the random variables $X_1, \ldots, X_n$ with a joint c.d.f. $F(x_1, \ldots, x_n)$, we can simply speak of the distribution of the random vector $\boldsymbol{X}$ with c.d.f. $F(\boldsymbol{x})$. When this vector notation is used, it must be kept in mind that if $\boldsymbol{X}$ is an n-dimensional random vector, then its c.d.f. is defined as a function on n-dimensional space R^n. At each point $\boldsymbol{x} = (x_1, \ldots, x_n) \in R^n$, the value of $F(\boldsymbol{x})$ is specified by Eq. (3.7.1).

Definition 3.7.2

Joint Discrete Distribution/p.f. It is said that n random variables $X_1, \ldots, X_n$ have a *discrete joint distribution* if the random vector $(X_1, \ldots, X_n)$ can have only a finite number or an infinite sequence of different possible values $(x_1, \ldots, x_n)$ in R^n. The *joint p.f.* of $X_1, \ldots, X_n$ is then defined as the function f such that for every point $(x_1, \ldots, x_n) \in R^n$,

$$f(x_1, \ldots, x_n) = \Pr(X_1 = x_1, \ldots, X_n = x_n).$$

In vector notation, Definition 3.7.2 says that the random vector $\boldsymbol{X}$ has a discrete distribution and that its p.f. is specified at every point $\boldsymbol{x} \in R^n$ by the relation

$$f(\boldsymbol{x}) = \Pr(\boldsymbol{X} = \boldsymbol{x}).$$

The following result is a simple generalization of Theorem 3.4.2.

Theorem 3.7.1 If $\boldsymbol{X}$ has a joint discrete distribution with joint p.f. f, then for every subset $C \subset R^n$,

$$\Pr(\boldsymbol{X} \in C) = \sum_{x \in C} f(\boldsymbol{x}).$$ ■

It is easy to show that, if each of $X_1, \ldots, X_n$ has a discrete distribution, then $\boldsymbol{X} = (X_1, \ldots, X_n)$ has a discrete joint distribution.

Example 3.7.3 A Clinical Trial. Consider the m patients in Example 3.7.1. Suppose for now that $P = p$ is known so that we don't treat it as a random variable. The joint p.f. of $\boldsymbol{X} = (X_1, \ldots, X_m)$ is

$$f(\boldsymbol{x}) = p^{x_1 + \cdots + x_m}(1-p)^{m - x_1 - \cdots - x_m},$$

for all $x_i \in \{0, 1\}$ and 0 otherwise. ◀

Definition 3.7.3 Continuous Distribution/p.d.f. It is said that n random variables $X_1, \ldots, X_n$ have a continuous joint distribution if there is a nonnegative function f defined on R^n such that for every subset $C \subset R^n$,

$$\Pr[(X_1, \ldots, X_n) \in C] = \int \cdots \int_C f(x_1, \ldots x_n)\, dx_1 \cdots dx_n, \tag{3.7.2}$$

if the integral exists. The function f is called the *joint p.d.f.* of $X_1, \ldots, X_n$.

In vector notation, $f(\boldsymbol{x})$ denotes the p.d.f. of the random vector $\boldsymbol{X}$ and Eq. (3.7.2) could be rewritten more simply in the form

$$\Pr(\boldsymbol{X} \in C) = \int \cdots \int_C f(\boldsymbol{x})\, d\boldsymbol{x}.$$

Theorem 3.7.2 If the joint distribution of $X_1, \ldots, X_n$ is continuous, then the joint p.d.f. f can be derived from the joint c.d.f. F by using the relation

$$f(x_1, \ldots, x_n) = \frac{\partial^n F(x_1, \ldots, x_n)}{\partial x_1 \cdots \partial x_n}$$

at all points $(x_1, \ldots, x_n)$ at which the derivative in this relation exists. ■

Example 3.7.4 Failure Times. We can find the joint p.d.f. for the three random variables in Example 3.7.2 by applying Theorem 3.7.2. The third-order mixed partial is easily calculated to be

$$f(x_1, x_2, x_3) = \begin{cases} 6e^{-x_1 - 2x_2 - 3x_3} & \text{for } x_1, x_2, x_3 > 0, \\ 0 & \text{otherwise.} \end{cases}$$ ◀

It is important to note that, even if each of $X_1, \ldots, X_n$ has a continuous distribution, the vector $\boldsymbol{X} = (X_1, \ldots, X_n)$ might not have a continuous joint distribution. See Exercise 9 in this section.

Example 3.7.5 Service Times in a Queue. A queue is a system in which customers line up for service and receive their service according to some algorithm. A simple model is the single-server queue, in which all customers wait for a single server to serve everyone ahead of them in the line and then they get served. Suppose that n customers arrive at a

single-server queue for service. Let X_i be the time that the server spends serving customer i for $i = 1, \ldots, n$. We might use a joint distribution for $\boldsymbol{X} = (X_1, \ldots, X_n)$ with joint p.d.f. of the form

$$f(\boldsymbol{x}) = \begin{cases} \dfrac{c}{\left(2 + \sum_{i=1}^{n} x_i\right)^{n+1}} & \text{for all } x_i > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{3.7.3}$$

We shall now find the value of c such that the function in Eq. (3.7.3) is a joint p.d.f. We can do this by integrating over each variable $x_1, \ldots, x_n$ in succession (starting with x_n). The first integral is

$$\int_0^\infty \frac{c}{(2 + x_1 + \cdots + x_n)^{n+1}}\, dx_n = \frac{c/n}{(2 + x_1 + \cdots + x_{n-1})^n}. \tag{3.7.4}$$

The right-hand side of Eq. (3.7.4) is in the same form as the original p.d.f. except that n has been reduced to $n - 1$ and c has been divided by n. It follows that when we integrate over the variable x_i (for $i = n - 1, n - 2, \ldots, 1$), the result will be in the same form with n reduced to $i - 1$ and c divided by $n(n - 1) \cdots i$. The result of integrating all coordinates except x_1 is then

$$\frac{c/n!}{(2 + x_1)^2}, \quad \text{for } x_1 > 0.$$

Integrating x_1 out of this yields $c/[2(n!)]$, which must equal 1, so $c = 2(n!)$. ◀

Mixed Distributions

Example 3.7.6 Arrivals at a Queue. In Example 3.7.5, we introduced the single-server queue and discussed service times. Some features that influence the performance of a queue are the rate at which customers arrive and the rate at which customers are served. Let Z stand for the rate at which customers are served, and let Y stand for the rate at which customers arrive at the queue. Finally, let W stand for the number of customers that arrive during one day. Then W is discrete while Y and Z could be continuous random variables. A possible joint p.f./p.d.f. for these three random variables is

$$f(y, z, w) = \begin{cases} 6e^{-3z-10y}(8y)^w/w! & \text{for } z, y > 0 \text{ and } w = 0, 1, \ldots, \\ 0 & \text{otherwise.} \end{cases}$$

We can verify this claim shortly. ◀

Definition 3.7.4 Joint p.f./p.d.f. Let $X_1, \ldots, X_n$ be random variables, some of which have a continuous joint distribution and some of which have discrete distributions; their joint distribution would then be represented by a function f that we call the *joint p.f./p.d.f.*. The function has the property that the probability that $\boldsymbol{X}$ lies in a subset $C \subset R^n$ is calculated by summing $f(\boldsymbol{x})$ over the values of the coordinates of $\boldsymbol{x}$ that correspond to the discrete random variables and integrating over those coordinates that correspond to the continuous random variables for all points $\boldsymbol{x} \in C$.

Example 3.7.7 Arrivals at a Queue. We shall now verify that the proposed p.f./p.d.f. in Example 3.7.6 actually sums and integrates to 1 over all values of (y, z, w). We must sum over w and integrate over y and z. We have our choice of in what order to do them. It is not

difficult to see that we can factor f as $f(y, z, w) = h_2(z)h_{13}(y, w)$, where

$$h_2(z) = \begin{cases} 6e^{-3z} & \text{for } z > 0, \\ 0 & \text{otherwise,} \end{cases}$$

$$h_{13}(y, w) = \begin{cases} e^{-10y}(8y)^w/w! & \text{for } y > 0 \text{ and } w = 0, 1, \ldots, \\ 0 & \text{otherwise.} \end{cases}$$

So we can integrate z out first to get

$$\int_{-\infty}^{\infty} f(y, z, w)dz = h_{13}(y, w) \int_0^{\infty} 6e^{-3z}dz = 2h_{13}(y, w).$$

Integrating y out of $h_{13}(y, w)$ is possible, but not pleasant. Instead, notice that $(8y)^w/w!$ is the wth term in the Taylor expansion of e^{8y}. Hence,

$$\sum_{w=0}^{\infty} 2h_{13}(y, w) = 2e^{-10y} \sum_{w=0}^{\infty} \frac{(8y)^w}{w!} = 2e^{-10y}e^{8y} = 2e^{-2y},$$

for $y > 0$ and 0 otherwise. Finally, integrating over y yields 1. ◀

Example 3.7.8

A Clinical Trial. In Example 3.7.1, one of the random variables P has a continuous distribution, and the others $X_1, \ldots, X_m$ have discrete distributions. A possible joint p.f./p.d.f. for $(X_1, \ldots, X_m, P)$ is

$$f(\boldsymbol{x}, p) = \begin{cases} p^{x_1+\cdots+x_m}(1-p)^{m-x_1-\cdots-x_m} & \text{for all } x_i \in \{0, 1\} \text{ and } 0 \le p \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

We can find probabilities based on this function. Suppose, for example, that we want the probability that there is exactly one success among the first two patients, that is, $\Pr(X_1 + X_2 = 1)$. We must integrate $f(\boldsymbol{x}, p)$ over p and sum over all values of $\boldsymbol{x}$ that have $x_1 + x_2 = 1$. For purposes of illustration, suppose that $m = 4$. First, factor out $p^{x_1+x_2}(1-p)^{2-x_1-x_2} = p(1-p)$, which yields

$$f(\boldsymbol{x}, p) = [p(1-p)]p^{x_3+x_4}(1-p)^{2-x_3-x_4},$$

for $x_3, x_4 \in \{0, 1\}$, $0 < p < 1$, and $x_1 + x_2 = 1$. Summing over x_3 yields

$$[p(1-p)]\left(p^{x_4}(1-p)^{1-x_4}(1-p) + pp^{x_4}(1-p)^{1-x_4}\right) = [p(1-p)]p^{x_4}(1-p)^{1-x_4}.$$

Summing this over x_4 gives $p(1-p)$. Next, integrate over p to get $\int_0^1 p(1-p)dp = 1/6$. Finally, note that there are two (x_1, x_2) vectors, $(1, 0)$ and $(0, 1)$, that have $x_1 + x_2 = 1$, so $\Pr(X_1 + X_2 = 1) = (1/6) + (1/6) = 1/3$. ◀

Marginal Distributions

Deriving a Marginal p.d.f. If the joint distribution of n random variables $X_1, \ldots, X_n$ is known, then the marginal distribution of each single random variable X_i can be derived from this joint distribution. For example, if the joint p.d.f. of $X_1, \ldots, X_n$ is f, then the marginal p.d.f. f_1 of X_1 is specified at every value x_1 by the relation

$$f_1(x_1) = \underbrace{\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty}}_{n-1} f(x_1, \ldots, x_n)\, dx_2 \cdots dx_n.$$

More generally, the marginal joint p.d.f. of any k of the n random variables $X_1, \ldots, X_n$ can be found by integrating the joint p.d.f. over all possible values of

the other $n - k$ variables. For example, if f is the joint p.d.f. of four random variables X_1, X_2, X_3, and X_4, then the marginal bivariate p.d.f. f_{24} of X_2 and X_4 is specified at each point (x_2, x_4) by the relation

$$f_{24}(x_2, x_4) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x_1, x_2, x_3, x_4)\, dx_1\, dx_3.$$

Example 3.7.9

Service Times in a Queue. Suppose that $n = 5$ in Example 3.7.5 and that we want the marginal bivariate p.d.f. of (X_1, X_4). We must integrate Eq. (3.7.3) over x_2, x_3, and x_5. Since the joint p.d.f. is symmetric with respect to permutations of the coordinates of $\boldsymbol{x}$, we shall just integrate over the last three variables and then change the names of the remaining variables to x_1 and x_4. We already saw how to do this in Example 3.7.5. The result is

$$f_{12}(x_1, x_2) = \begin{cases} \dfrac{4}{(2 + x_1 + x_2)^3} & \text{for } x_1, x_2 > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{3.7.5}$$

Then f_{14} is just like (3.7.5) with all the 2 subscripts changed to 4. The univariate marginal p.d.f. of each X_i is

$$f_i(x_i) = \begin{cases} \dfrac{2}{(2 + x_i)^2} & \text{for } x_i > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{3.7.6}$$

So, for example, if we want to know how likely it is that a customer will have to wait longer than three time units, we can calculate $\Pr(X_i > 3)$ by integrating the function in Eq. (3.7.6) from 3 to ∞. The result is 0.4. ◀

If n random variables $X_1, \ldots, X_n$ have a discrete joint distribution, then the marginal joint p.f. of each subset of the n variables can be obtained from relations similar to those for continuous distributions. In the new relations, the integrals are replaced by sums.

Deriving a Marginal c.d.f. Consider now a joint distribution for which the joint c.d.f. of $X_1, \ldots, X_n$ is F. The marginal c.d.f. F_1 of X_1 can be obtained from the following relation:

$$\begin{aligned} F_1(x_1) &= \Pr(X_1 \le x_1) = \Pr(X_1 \le x_1, X_2 < \infty, \ldots, X_n < \infty) \\ &= \lim_{x_2, \ldots, x_n \to \infty} F(x_1, x_2, \ldots, x_n). \end{aligned}$$

Example 3.7.10

Failure Times. We can find the marginal c.d.f. of X_1 from the joint c.d.f. in Example 3.7.2 by letting x_2 and x_3 go to ∞. The limit is $F_1(x_1) = 1 - e^{-x_1}$ for $x_1 \ge 0$ and 0 otherwise. ◀

More generally, the marginal joint c.d.f. of any k of the n random variables $X_1, \ldots, X_n$ can be found by computing the limiting value of the n-dimensional c.d.f. F as $x_j \to \infty$ for each of the other $n - k$ variables x_j. For example, if F is the joint c.d.f. of four random variables X_1, X_2, X_3, and X_4, then the marginal bivariate c.d.f. F_{24} of X_2 and X_4 is specified at every point (x_2, x_4) by the relation

$$F_{24}(x_2, x_4) = \lim_{x_1, x_3 \to \infty} F(x_1, x_2, x_3, x_4).$$

Example 3.7.11 Failure Times. We can find the marginal bivariate c.d.f. of X_1 and X_3 from the joint c.d.f. in Example 3.7.2 by letting x_2 go to ∞. The limit is

$$F_{13}(x_1, x_3) = \begin{cases} (1 - e^{-x_1})(1 - e^{-3x_3}) & \text{for } x_1, x_3 \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$ ◀

Independent Random Variables

Definition 3.7.5 Independent Random Variables. It is said that n random variables $X_1, \ldots, X_n$ are *independent* if, for every n sets $A_1, A_2, \ldots, A_n$ of real numbers,

$$\Pr(X_1 \in A_1, X_2 \in A_2, \ldots, X_n \in A_n) = \Pr(X_1 \in A_1) \Pr(X_2 \in A_2) \cdots \Pr(X_n \in A_n).$$

If $X_1, \ldots, X_n$ are independent, it follows easily that the random variables in every nonempty subset of $X_1, \ldots, X_n$ are also independent. (See Exercise 11.)

There is a generalization of Theorem 3.5.4.

Theorem 3.7.3 Let F denote the joint c.d.f. of $X_1, \ldots, X_n$, and let F_i denote the marginal univariate c.d.f. of X_i for $i = 1, \ldots, n$. The variables $X_1, \ldots, X_n$ are independent if and only if, for all points $(x_1, x_2, \ldots, x_n) \in R^n$,

$$F(x_1, x_2, \ldots, x_n) = F_1(x_1) F_2(x_2) \cdots F_n(x_n).$$ ■

Theorem 3.7.3 says that $X_1, \ldots, X_n$ are independent if and only if their joint c.d.f. is the product of their n individual marginal c.d.f.'s. It is easy to check that the three random variables in Example 3.7.2 are independent using Theorem 3.7.3.

There is also a generalization of Corollary 3.5.1.

Theorem 3.7.4 If $X_1, \ldots, X_n$ have a continuous, discrete, or mixed joint distribution for which the joint p.d.f., joint p.f., or joint p.f./p.d.f. is f, and if f_i is the marginal univariate p.d.f. or p.f. of X_i $(i = 1, \ldots, n)$, then $X_1, \ldots, X_n$ are independent if and only if the following relation is satisfied at all points $(x_1, x_2, \ldots, x_n) \in R^n$:

$$f(x_1, x_2, \ldots, x_n) = f_1(x_1) f_2(x_2) \cdots f_n(x_n). \tag{3.7.7}$$ ■

Example 3.7.12 Service Times in a Queue. In Example 3.7.9, we can multiply together the two univariate marginal p.d.f.'s of X_1 and X_2 calculated using Eq. (3.7.6) and see that the product does *not* equal the bivariate marginal p.d.f. of (X_1, X_2) in Eq. (3.7.5). So X_1 and X_2 are not independent. ◀

Definition 3.7.6 Random Samples/i.i.d./Sample Size. Consider a given probability distribution on the real line that can be represented by either a p.f. or a p.d.f. f. It is said that n random variables $X_1, \ldots, X_n$ form a *random sample* from this distribution if these random variables are independent and the marginal p.f. or p.d.f. of each of them is f. Such random variables are also said to be *independent and identically distributed*, abbreviated *i.i.d.* We refer to the number n of random variables as the *sample size*.

Definition 3.7.6 says that $X_1, \ldots, X_n$ form a random sample from the distribution represented by f if their joint p.f. or p.d.f. g is specified as follows at all points $(x_1, x_2, \ldots, x_n) \in R^n$:

$$g(x_1, \ldots, x_n) = f(x_1) f(x_2) \cdots f(x_n).$$

Clearly, an i.i.d. sample cannot have a mixed joint distribution.

Example 3.7.13 Lifetimes of Light Bulbs. Suppose that the lifetime of each light bulb produced in a certain factory is distributed according to the following p.d.f.:

$$f(x) = \begin{cases} xe^{-x} & \text{for } x > 0, \\ 0 & \text{otherwise.} \end{cases}$$

We shall determine the joint p.d.f. of the lifetimes of a random sample of n light bulbs drawn from the factory's production.

The lifetimes $X_1, \ldots, X_n$ of the selected bulbs will form a random sample from the p.d.f. f. For typographical simplicity, we shall use the notation $\exp(v)$ to denote the exponential e^v when the expression for v is complicated. Then the joint p.d.f. g of $X_1, \ldots, X_n$ will be as follows: If $x_i > 0$ for $i = 1, \ldots, n$,

$$\begin{aligned} g(x_1, \ldots, x_n) &= \prod_{i=1}^{n} f(x_i) \\ &= \left(\prod_{i=1}^{n} x_i\right) \exp\left(-\sum_{i=1}^{n} x_i\right). \end{aligned}$$

Otherwise, $g(x_1, \ldots, x_n) = 0$.

Every probability involving the n lifetimes $X_1, \ldots, X_n$ can in principle be determined by integrating this joint p.d.f. over the appropriate subset of R^n. For example, if C is the subset of points $(x_1, \ldots, x_n)$ such that $x_i > 0$ for $i = 1, \ldots, n$ and $\sum_{i=1}^{n} x_i < a$, where a is a given positive number, then

$$\Pr\left(\sum_{i=1}^{n} X_i < a\right) = \int \cdots \int_C \left(\prod_{i=1}^{n} x_i\right) \exp\left(-\sum_{i=1}^{n} x_i\right) dx_1 \cdots dx_n. \quad \blacktriangleleft$$

The evaluation of the integral given at the end of Example 3.7.13 may require a considerable amount of time without the aid of tables or a computer. Certain other probabilities, however, can be evaluated easily from the basic properties of continuous distributions and random samples. For example, suppose that for the conditions of Example 3.7.13 it is desired to find $\Pr(X_1 < X_2 < \cdots < X_n)$. Since the random variables $X_1, \ldots, X_n$ have a continuous joint distribution, the probability that at least two of these random variables will have the same value is 0. In fact, the probability is 0 that the vector $(X_1, \ldots, X_n)$ will belong to each specific subset of R^n for which the n-dimensional volume is 0. Furthermore, since $X_1, \ldots, X_n$ are independent and identically distributed, each of these variables is equally likely to be the smallest of the n lifetimes, and each is equally likely to be the largest. More generally, if the lifetimes $X_1, \ldots, X_n$ are arranged in order from the smallest to the largest, each particular ordering of $X_1, \ldots, X_n$ is as likely to be obtained as any other ordering. Since there are $n!$ different possible orderings, the probability that the particular ordering $X_1 < X_2 < \cdots < X_n$ will be obtained is $1/n!$. Hence,

$$\Pr(X_1 < X_2 < \cdots < X_n) = \frac{1}{n!}.$$

Conditional Distributions

Suppose that n random variables $X_1, \ldots, X_n$ have a continuous joint distribution for which the joint p.d.f. is f and that f_0 denotes the marginal joint p.d.f. of the $k < n$ random variables $X_1, \ldots, X_k$. Then for all values of $x_1, \ldots, x_k$ such that $f_0(x_1, \ldots, x_k) > 0$, the conditional p.d.f. of $(X_{k+1}, \ldots, X_n)$ given that $X_1 = x_1, \ldots, X_k = x_k$ is defined

as follows:

$$g_{k+1\ldots n}(x_{k+1}, \ldots, x_n | x_1, \ldots, x_k) = \frac{f(x_1, x_2, \ldots, x_n)}{f_0(x_1, \ldots, x_k)}.$$

The definition above generalizes to arbitrary joint distributions as follows.

Definition 3.7.7

Conditional p.f., p.d.f., or p.f./p.d.f. Suppose that the random vector $\boldsymbol{X} = (X_1, \ldots, X_n)$ is divided into two subvectors $\boldsymbol{Y}$ and $\boldsymbol{Z}$, where $\boldsymbol{Y}$ is a k-dimensional random vector comprising k of the n random variables in $\boldsymbol{X}$, and $\boldsymbol{Z}$ is an $(n-k)$-dimensional random vector comprising the other $n-k$ random variables in $\boldsymbol{X}$. Suppose also that the n-dimensional joint p.f., p.d.f., or p.f./p.d.f. of $(\boldsymbol{Y}, \boldsymbol{Z})$ is f and that the marginal $(n-k)$-dimensional p.f., p.d.f., or p.f./p.d.f. of $\boldsymbol{Z}$ is f_2. Then for every given point $\boldsymbol{z} \in R^{n-k}$ such that $f_2(\boldsymbol{z}) > 0$, the conditional k-dimensional p.f., p.d.f., or p.f./p.d.f. g_1 of $\boldsymbol{Y}$ given $\boldsymbol{Z} = \boldsymbol{z}$ is defined as follows:

$$g_1(\boldsymbol{y}|\boldsymbol{z}) = \frac{f(\boldsymbol{y}, \boldsymbol{z})}{f_2(\boldsymbol{z})} \qquad \text{for } \boldsymbol{y} \in R^k. \tag{3.7.8}$$

Eq. (3.7.8) can be rewritten as

$$f(\boldsymbol{y}, \boldsymbol{z}) = g_1(\boldsymbol{y}|\boldsymbol{z}) f_2(\boldsymbol{z}), \tag{3.7.9}$$

which allows construction of the joint distribution from a conditional distribution and a marginal distribution. As in the bivariate case, it is safe to assume that $f(\boldsymbol{y}, \boldsymbol{z}) = 0$ whenever $f_2(\boldsymbol{z}) = 0$. Then Eq. (3.7.9) holds for all $\boldsymbol{y}$ and $\boldsymbol{z}$ even though $g_1(\boldsymbol{y}|\boldsymbol{z})$ is not uniquely defined.

Example 3.7.14

Service Times in a Queue. In Example 3.7.9, we calculated the marginal bivariate distribution of two service times $\boldsymbol{Z} = (X_1, X_2)$. We can now find the conditional three-dimensional p.d.f. of $\boldsymbol{Y} = (X_3, X_4, X_5)$ given $\boldsymbol{Z} = (x_1, x_2)$ for every pair (x_1, x_2) such that $x_1, x_2 > 0$:

$$\begin{aligned} g_1(x_3, x_4, x_5 | x_1, x_2) &= \frac{f(x_1, \ldots, x_5)}{f_{12}(x_1, x_2)} \\ &= \left(\frac{240}{(2 + x_1 + \cdots + x_5)^6}\right)\left(\frac{4}{(2 + x_1 + x_2)^3}\right)^{-1} \\ &= \frac{60(2 + x_1 + x_2)^3}{(2 + x_1 + \cdots + x_5)^6}, \end{aligned} \tag{3.7.10}$$

for $x_3, x_4, x_5 > 0$, and 0 otherwise. The joint p.d.f. in (3.7.10) looks like a bunch of symbols, but it can be quite useful. Suppose that we observe $X_1 = 4$ and $X_2 = 6$. Then

$$g_1(x_3, x_4, x_5 | 4.6) = \begin{cases} \dfrac{103{,}680}{(12 + x_3 + x_4 + x_5)^6} & \text{for } x_3, x_4, x_5 > 0, \\ 0 & \text{otherwise.} \end{cases}$$

We can now calculate the conditional probability that $X_3 > 3$ given $X_1 = 4$, $X_2 = 6$:

$$\Pr(X_3 > 3|X_1 = 4,\ X_2 = 6) = \int_3^\infty \int_0^\infty \int_0^\infty \frac{10{,}360}{(12 + x_3 + x_4 + x_5)^6} dx_5 dx_4 dx_3$$
$$= \int_3^\infty \int_0^\infty \frac{20{,}736}{(12 + x_3 + x_4)^5} dx_4 dx_3$$
$$= \int_3^\infty \frac{5184}{(12 + x_3)^4} dx_3$$
$$= \frac{1728}{15^3} = 0.512.$$

Compare this to the calculation of $\Pr(X_3 > 3) = 0.4$ at the end of Example 3.7.9. After learning that the first two service times are a bit longer than three time units, we revise the probability that $X_3 > 3$ upward to reflect what we learned from the first two observations. If the first two service times had been small, the conditional probability that $X_3 > 3$ would have been smaller than 0.4. For example, $\Pr(X_3 > 3|X_1 = 1,\ X_2 = 1.5) = 0.216$. ◀

Example 3.7.15 Determining a Marginal Bivariate p.d.f. Suppose that Z is a random variable for which the p.d.f. f_0 is as follows:

$$f_0(z) = \begin{cases} 2e^{-2z} & \text{for } z > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{3.7.11}$$

Suppose, furthermore, that for every given value $Z = z > 0$ two other random variables X_1 and X_2 are independent and identically distributed and the conditional p.d.f. of each of these variables is as follows:

$$g(x|z) = \begin{cases} ze^{-zx} & \text{for } x > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{3.7.12}$$

We shall determine the marginal joint p.d.f. of (X_1, X_2).

Since X_1 and X_2 are i.i.d. for each given value of Z, their conditional joint p.d.f. when $Z = z > 0$ is

$$g_{12}(x_1, x_2|z) = \begin{cases} z^2 e^{-z(x_1+x_2)} & \text{for } x_1, x_2 > 0, \\ 0 & \text{otherwise.} \end{cases}$$

The joint p.d.f. f of (Z, X_1, X_2) will be positive only at those points (z, x_1, x_2) such that $x_1, x_2, z > 0$. It now follows that, at every such point,

$$f(z, x_1, x_2) = f_0(z) g_{12}(x_1, x_2|z) = 2z^2 e^{-z(2+x_1+x_2)}.$$

For $x_1 > 0$ and $x_2 > 0$, the marginal joint p.d.f. $f_{12}(x_1, x_2)$ of X_1 and X_2 can be determined either using integration by parts or some special results that will arise in Sec. 5.7:

$$f_{12}(x_1, x_2) = \int_0^\infty f(z, x_1, x_2)\, dz = \frac{4}{(2 + x_1 + x_2)^3},$$

for $x_1, x_2 > 0$. The reader will note that this p.d.f. is the same as the marginal bivariate p.d.f. of (X_1, X_2) found in Eq. (3.7.5).

From this marginal bivariate p.d.f., we can evaluate probabilities involving X_1 and X_2, such as $\Pr(X_1 + X_2 < 4)$. We have

$$\Pr(X_1 + X_2 < 4) = \int_0^4 \int_0^{4-x_2} \frac{4}{(2 + x_1 + x_2)^3}\, dx_1\, dx_2 = \frac{4}{9}.$$ ◀

Example 3.7.16 Service Times in a Queue. We can think of the random variable Z in Example 3.7.15 as the rate at which customers are served in the queue of Example 3.7.5. With this interpretation, it is useful to find the conditional distribution of the rate Z after we observe some of the service times such as X_1 and X_2.

For every value of z, the conditional p.d.f. of Z given $X_1 = x_1$ and $X_2 = x_2$ is

$$g_0(z|x_1, x_2) = \frac{f(z, x_1, x_2)}{f_{12}(x_1, x_2)} = \begin{cases} \frac{1}{2}(2 + x_1 + x_2)^3 z^2 e^{-z(2+x_1+x_2)} & \text{for } z > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{3.7.13}$$

Finally, we shall evaluate $\Pr(Z \le 1|X_1 = 1,\ X_2 = 4)$. We have

$$\Pr(Z \le 1|X_1 = 1,\ X_2 = 4) = \int_0^1 g_0(z|1, 4)\, dz = \int_0^1 171.5 z^2 e^{-7z}\, dz = 0.9704. \quad \blacktriangleleft$$

Law of Total Probability and Bayes' Theorem Example 3.7.15 contains an example of the multivariate version of the law of total probability, while Example 3.7.16 contains an example of the multivariate version of Bayes' theorem. The proofs of the general versions are straightforward consequences of Definition 3.7.7.

Theorem 3.7.5 Multivariate Law of Total Probability and Bayes' Theorem. Assume the conditions and notation given in Definition 3.7.7. If $\boldsymbol{Z}$ has a continuous joint distribution, the marginal p.d.f. of $\boldsymbol{Y}$ is

$$f_1(\boldsymbol{y}) = \underbrace{\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty}}_{n-k} g_1(\boldsymbol{y}|\boldsymbol{z})\, f_2(\boldsymbol{z})\, d\boldsymbol{z}, \tag{3.7.14}$$

and the conditional p.d.f. of $\boldsymbol{Z}$ given $\boldsymbol{Y} = \boldsymbol{y}$ is

$$g_2(\boldsymbol{z}|\boldsymbol{y}) = \frac{g_1(\boldsymbol{y}|\boldsymbol{z})\, f_2(\boldsymbol{z})}{f_1(\boldsymbol{y})}. \tag{3.7.15}$$

If $\boldsymbol{Z}$ has a discrete joint distribution, then the multiple integral in (3.7.14) must be replaced by a multiple summation. If $\boldsymbol{Z}$ has a mixed joint distribution, the multiple integral must be replaced by integration over those coordinates with continuous distributions and summation over those coordinates with discrete distributions. ■

Conditionally Independent Random Variables In Examples 3.7.15 and 3.7.16, $\boldsymbol{Z}$ is the single random variable Z and $\boldsymbol{Y} = (X_1, X_2)$. These examples also illustrate the use of conditionally independent random variables. That is, X_1 and X_2 are conditionally independent given $Z = z$ for all $z > 0$. In Example 3.7.16, we said that Z was the rate at which customers were served. When this rate is unknown, it is a major source of uncertainty. Partitioning the sample space by the values of the rate Z and then conditioning on each value of Z removes a major source of uncertainty for part of the calculation.

In general, conditional independence for random variables is similar to conditional independence for events.

Definition 3.7.8 Conditionally Independent Random Variables. Let $\boldsymbol{Z}$ be a random vector with joint p.f., p.d.f., or p.f./p.d.f. $f_0(\boldsymbol{z})$. Several random variables $X_1, \ldots, X_n$ are *conditionally independent given* $\boldsymbol{Z}$ if, for all $\boldsymbol{z}$ such that $f_0(\boldsymbol{z}) > 0$, we have

$$g(\boldsymbol{x}|\boldsymbol{z}) = \prod_{i=1}^{n} g_i(x_i|\boldsymbol{z}),$$

where $g(\boldsymbol{x}|\boldsymbol{z})$ stands for the conditional multivariate p.f., p.d.f., or p.f./p.d.f. of $\boldsymbol{X}$ given $\boldsymbol{Z} = \boldsymbol{z}$ and $g_i(x_i|\boldsymbol{z})$ stands for the conditional univariate p.f. or p.d.f. of X_i given $\boldsymbol{Z} = \boldsymbol{z}$.

In Example 3.7.15, $g_i(x_i|z) = ze^{-zx_i}$ for $x_i > 0$ and $i = 1, 2$.

Example 3.7.17 A Clinical Trial. In Example 3.7.8, the joint p.f./p.d.f. given there was constructed by assuming that $X_1, \ldots, X_m$ were conditionally independent given $P = p$ each with the same conditional p.f., $g_i(x_i|p) = p^{x_i}(1-p)^{1-x_i}$ for $x_i \in \{0, 1\}$ and that P had the uniform distribution on the interval $[0, 1]$. These assumptions produce, in the notation of Definition 3.7.8,

$$g(\boldsymbol{x}|p) = \begin{cases} p^{x_1+\cdots+x_m}(1-p)^{40-x_1-\cdots-x_m} & \text{for all } x_i \in \{0, 1\} \text{ and } 0 \le p \le 1, \\ 0 & \text{otherwise,} \end{cases}$$

for $0 \le p \le 1$. Combining this with the marginal p.d.f. of P, $f_2(p) = 1$ for $0 \le p \le 1$ and 0 otherwise, we get the joint p.f./p.d.f. given in Example 3.7.8. ◀

Conditional Versions of Past and Future Theorems We mentioned earlier that conditional distributions behave just like distributions. Hence, all theorems that we have proven and will prove in the future have conditional versions. For example, the law of total probability in Eq. (3.7.14) has the following version conditional on another random vector $\boldsymbol{W} = \boldsymbol{w}$:

$$f_1(\boldsymbol{y}|\boldsymbol{w}) = \underbrace{\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty}}_{n-k} g_1(\boldsymbol{y}|\boldsymbol{z}, \boldsymbol{w}) f_2(\boldsymbol{z}|\boldsymbol{w})\, d\boldsymbol{z}, \tag{3.7.16}$$

where $f_1(\boldsymbol{y}|\boldsymbol{w})$ stands for the conditional p.d.f., p.f., or p.f./p.d.f. of $\boldsymbol{Y}$ given $\boldsymbol{W} = \boldsymbol{w}$, $g_1(\boldsymbol{y}|\boldsymbol{z}, \boldsymbol{w})$ stands for the conditional p.d.f., p.f., or p.f./p.d.f. of $\boldsymbol{Y}$ given $(\boldsymbol{Z}, \boldsymbol{W}) = (\boldsymbol{z}, \boldsymbol{w})$, and $f_2(\boldsymbol{z}|\boldsymbol{w})$ stands for the conditional p.d.f. of $\boldsymbol{Z}$ given $\boldsymbol{W} = \boldsymbol{w}$. Using the same notation, the conditional version of Bayes' theorem is

$$g_2(\boldsymbol{z}|\boldsymbol{y}, \boldsymbol{w}) = \frac{g_1(\boldsymbol{y}|\boldsymbol{z}, \boldsymbol{w}) f_2(\boldsymbol{z}|\boldsymbol{w})}{f_1(\boldsymbol{y}|\boldsymbol{w})}. \tag{3.7.17}$$

Example 3.7.18 Conditioning on Random Variables in Sequence. In Example 3.7.15, we found the conditional p.d.f. of Z given $(X_1, X_2) = (x_1, x_2)$. Suppose now that there are three more observations available, X_3, X_4, and X_5, and suppose that all of $X_1, \ldots, X_5$ are conditionally i.i.d. given $Z = z$ with p.d.f. $g(x|z)$. We shall use the conditional version of Bayes' theorem to compute the conditional p.d.f. of Z given $(X_1, \ldots, X_5) = (x_1, \ldots, x_5)$. First, we shall find the conditional p.d.f. $g_{345}(x_3, x_4, x_5|x_1, x_2, z)$ of $\boldsymbol{Y} = (X_3, X_4, X_5)$ given $Z = z$ and $\boldsymbol{W} = (X_1, X_2) = (x_1, x_2)$. We shall use the notation for p.d.f.'s in the discussion immediately preceding this example. Since $X_1, \ldots, X_5$ are conditionally i.i.d. given Z, we have that $g_1(\boldsymbol{y}|z, \boldsymbol{w})$ does not depend on $\boldsymbol{w}$. In fact,

$$g_1(\boldsymbol{y}|z, \boldsymbol{w}) = g(x_3|z)g(x_4|z)g(x_5|z) = z^3 e^{-z(x_3+x_4+x_5)},$$

for $x_3, x_4, x_5 > 0$. We also need the conditional p.d.f. of Z given $\boldsymbol{W} = \boldsymbol{w}$, which was calculated in Eq. (3.7.13), and we now denote it

$$f_2(z|\boldsymbol{w}) = \frac{1}{2}(2 + x_1 + x_2)^3 z^2 e^{-z(2+x_1+x_2)}.$$

Finally, we need the conditional p.d.f. of the last three observations given the first two. This was calculated in Example 3.7.14, and we now denote it

$$f_1(\boldsymbol{y}|\boldsymbol{w}) = \frac{60(2 + x_1 + x_2)^3}{(2 + x_1 + \cdots + x_5)^6}.$$

Now combine these using Bayes' theorem (3.7.17) to obtain

$$g_2(\boldsymbol{z}|\boldsymbol{y}, \boldsymbol{w}) = \frac{z^3 e^{-z(x_3+x_4+x_5)} \frac{1}{2}(2 + x_1 + x_2)^3 z^2 e^{-z(2+x_1+x_2)}}{\frac{60(2 + x_1 + x_2)^3}{(2 + x_1 + \cdots + x_5)^6}}$$

$$= \frac{1}{120}(2 + x_1 + \cdots + x_5)^6 z^5 e^{-z(2+x_1+\cdots+x_5)},$$

for $z > 0$. ◀

Note: Simple Rule for Creating Conditional Versions of Results. If you ever wish to determine the conditional version given $\boldsymbol{W} = \boldsymbol{w}$ of a result that you have proven, here is a simple method. Just add "conditional on $\boldsymbol{W} = \boldsymbol{w}$" to every probabilistic statement in the result. This includes all probabilities, c.d.f.'s, quantiles, names of distributions, p.d.f.'s, p.f.'s, and so on. It also includes all future probabilistic concepts that we introduce in later chapters (such as expected values and variances in Chapter 4).

Note: Independence is a Special Case of Conditional Independence. Let $X_1, \ldots, X_n$ be independent random variables, and let W be a constant random variable. That is, there is a constant c such that $\Pr(W = c) = 1$. Then $X_1, \ldots, X_n$ are also conditionally independent given $W = c$. The proof is straightforward and is left to the reader (Exercise 15). This result is not particularly interesting in its own right. Its value is the following: If we prove a result for conditionally independent random variables or conditionally i.i.d. random variables, then the same result will hold for independent random variables or i.i.d. random variables as the case may be.

Histograms

Example 3.7.19

Rate of Service. In Examples 3.7.5 and 3.7.6, we considered customers arriving at a queue and being served. Let Z stand for the rate at which customers were served, and we let $X_1, X_2, \ldots$ stand for the times that the successive customers requrired for service. Assume that $X_1, X_2, \ldots$ are conditionally i.i.d. given $Z = z$ with p.d.f.

$$g(x|z) = \begin{cases} ze^{-zx} & \text{for } x > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{3.7.18}$$

This is the same as (3.7.12) from Example 3.7.15. In that example, we modeled Z as a random variable with p.d.f. $f_0(z) = 2\exp(-2z)$ for $z > 0$. In this example, we shall assume that $X_1, \ldots, X_n$ will be observed for some large value n, and we want to think about what these observations tell us about Z. To be specific, suppose that we observe $n = 100$ service times. The first 10 times are listed here:

1.39, 0.61, 2.47, 3.35, 2.56, 3.60, 0.32, 1.43, 0.51, 0.94.

The smallest and largest observed service times from the entire sample are 0.004 and 9.60, respectively. It would be nice to have a graphical display of the entire sample of $n = 100$ service times without having to list them separately. ◀

The histogram, defined below, is a graphical display of a collection of numbers. It is particularly useful for displaying the observed values of a collection of random variables that have been modeled as conditionally i.i.d.

Definition 3.7.9

Histogram. Let $x_1, \ldots, x_n$ be a collection of numbers that all lie between two values $a < b$. That is, $a \le x_i \le b$ for all $i = 1, \ldots, n$. Choose some integer $k \ge 1$ and divide the interval $[a, b]$ into k equal-length subintervals of length $(b - a)/k$. For each subinterval, count how many of the numbers $x_1, \ldots, x_n$ are in the subinterval. Let c_i be the count for subinterval i for $i = 1, \ldots, k$. Choose a number $r > 0$. (Typically, $r = 1$ or $r = n$ or $r = n(b - a)/k$.) Draw a two-dimensional graph with the horizonal axis running from a to b. For each subinterval $i = 1, \ldots, k$ draw a rectangular bar of width $(b - a)/k$ and height equal to c_i/r over the midpoint of the ith interval. Such a graph is called a *histogram*.

The choice of the number r in the definition of histogram depends on what one wishes to be displayed on the vertical axis. The shape of the histogram is identical regardless of what value one chooses for r. With $r = 1$, the height of each bar is the raw count for each subinterval, and counts are displayed on the vertical axis. With $r = n$, the height of each bar is the proportion of the set of numbers in each subinterval, and the vertical axis displays proportions. With $r = n(b - a)/k$, the area of each bar is the proportion of the set of numbers in each subinterval.

Example 3.7.20

Rate of Service. The $n = 100$ observed service times in Example 3.7.19 all lie between 0 and 10. It is convenient, in this example, to draw a histogram with horizontal axis running from 0 to 10 and divided into 10 subintervals of length 1 each. Other choices are possible, but this one will do for illustration. Figure 3.22 contains the histogram of the 100 observed service times with $r = 100$. One sees that the numbers of observed service times in the subintervals decrease as the center of the subinterval increses. This matches the behavior of the conditional p.d.f. $g(x|z)$ of the service times as a function of x for fixed z. ◀

Histograms are useful as more than just graphical displays of large sets of numbers. After we see the law of large numbers (Theorem 6.2.4), we can show that the

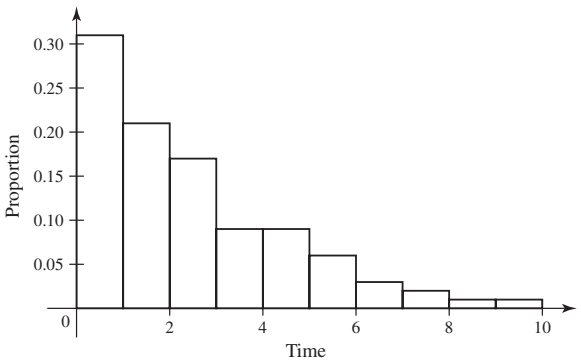


Figure 3.22 Histogram of service times for Example 3.7.20 with $a = 0$, $b = 10$, $k = 10$, and $r = 100$.

histogram of a large (conditionally) i.i.d. sample of continuous random variables is an approximation to the (conditional) p.d.f. of the random variables in the sample, so long as one uses the third choice of r, namely, $r = n(b-a)/k$.

Note: More General Histograms. Sometimes it is convenient to divide the range of the numbers to be plotted in a histogram into unequal-length subintervals. In such a case, one would typically let the height of each bar be c_i/r_i, where c_i is the raw count and r_i is proportional to the length of the ith subinterval. In this way, the area of each bar is still proportional to the count or proportion in each subinterval.

Summary

A finite collection of random variables is called a random vector. We have defined joint distributions for arbitrary random vectors. Every random vector has a joint c.d.f. Continuous random vectors have a joint p.d.f. Discrete random vectors have a joint p.f. Mixed distribution random vectors have a joint p.f./p.d.f. The coordinates of an n-dimensional random vector $\boldsymbol{X}$ are independent if the joint p.f., p.d.f., or p.f./p.d.f. $f(\boldsymbol{x})$ factors into $\prod_{i=1}^{n} f_i(x_i)$.

We can compute marginal distributions of subvectors of a random vector, and we can compute the conditional distribution of one subvector given the rest of the vector. We can construct a joint distribution for a random vector by piecing together a marginal distribution for part of the vector and a conditional distribution for the rest given the first part. There are versions of Bayes' theorem and the law of total probability for random vectors.

An n-dimensional random vector $\boldsymbol{X}$ has coordinates that are conditionally independent given $\boldsymbol{Z}$ if the conditional p.f., p.d.f., or p.f./p.d.f. $g(\boldsymbol{x}|\boldsymbol{z})$ of $\boldsymbol{X}$ given $\boldsymbol{Z} = \boldsymbol{z}$ factors into $\prod_{i=1}^{n} g_i(x_i|\boldsymbol{z})$. There are versions of Bayes' theorem, the law of total probability, and all future theorems about random variables and random vectors conditional on an arbitrary additional random vector.

Exercises

1. Suppose that three random variables X_1, X_2, and X_3 have a continuous joint distribution with the following joint p.d.f.: $f(x_1, x_2, x_3) =$

$$\begin{cases} c(x_1 + 2x_2 + 3x_3) & \text{for } 0 \le x_i \le 1 \ \ (i = 1, 2, 3), \\ 0 & \text{otherwise.} \end{cases}$$

Determine **(a)** the value of the constant c;
(b) the marginal joint p.d.f. of X_1 and X_3; and
(c) $\Pr\left(X_3 < \frac{1}{2} \middle| X_1 = \frac{1}{4},\ X_2 = \frac{3}{4}\right)$.

2. Suppose that three random variables X_1, X_2, and X_3 have a mixed joint distribution with p.f./p.d.f.:

$$f(x_1, x_2, x_3) = \begin{cases} c x_1^{1+x_2+x_3}(1-x_1)^{3-x_2-x_3} & \text{if } 0 < x_1 < 1 \\ & \quad \text{and } x_2, x_3 \in \{0, 1\}, \\ 0 & \text{otherwise.} \end{cases}$$

(Notice that X_1 has a continuous distribution and X_2 and X_3 have discrete distributions.) Determine **(a)** the value of the constant c; **(b)** the marginal joint p.f. of X_2 and X_3; and **(c)** the conditional p.d.f. of X_1 given $X_2 = 1$ and $X_3 = 1$.

3. Suppose that three random variables X_1, X_2, and X_3 have a continuous joint distribution with the following joint p.d.f.: $f(x_1, x_2, x_3) =$

$$\begin{cases} c e^{-(x_1+2x_2+3x_3)} & \text{for } x_i > 0 \ \ (i = 1, 2, 3), \\ 0 & \text{otherwise.} \end{cases}$$

Determine **(a)** the value of the constant c; **(b)** the marginal joint p.d.f. of X_1 and X_3; and **(c)** $\Pr(X_1 < 1 | X_2 = 2,\ X_3 = 1)$.

4. Suppose that a point (X_1, X_2, X_3) is chosen at random, that is, in accordance with the uniform p.d.f., from the following set S:

$$S = \{(x_1, x_2, x_3): 0 \le x_i \le 1 \text{ for } i = 1, 2, 3\}.$$

Determine:

a. $\Pr\left[\left(X_1 - \frac{1}{2}\right)^2 + \left(X_2 - \frac{1}{2}\right)^2 + \left(X_3 - \frac{1}{2}\right)^2 \le \frac{1}{4}\right]$

b. $\Pr(X_1^2 + X_2^2 + X_3^2 \le 1)$

5. Suppose that an electronic system contains n components that function independently of each other and that the probability that component i will function properly is p_i $(i = 1, \ldots, n)$. It is said that the components are connected *in series* if a necessary and sufficient condition for the system to function properly is that all n components function properly. It is said that the components are connected *in parallel* if a necessary and sufficient condition for the system to function properly is that at least one of the n components functions properly. The probability that the system will function properly is called the *reliability* of the system. Determine the reliability of the system, **(a)** assuming that the components are connected in series, and **(b)** assuming that the components are connected in parallel.

6. Suppose that the n random variables $X_1 \ldots, X_n$ form a random sample from a discrete distribution for which the p.f. is f. Determine the value of $\Pr(X_1 = X_2 = \cdots = X_n)$.

7. Suppose that the n random variables $X_1, \ldots, X_n$ form a random sample from a continuous distribution for which the p.d.f. is f. Determine the probability that at least k of these n random variables will lie in a specified interval $a \le x \le b$.

8. Suppose that the p.d.f. of a random variable X is as follows:

$$f(x) = \begin{cases} \frac{1}{n!}x^n e^{-x} & \text{for } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Suppose also that for any given value $X = x$ $(x > 0)$, the n random variables $Y_1, \ldots, Y_n$ are i.i.d. and the conditional p.d.f. g of each of them is as follows:

$$g(y|x) = \begin{cases} \frac{1}{x} & \text{for } 0 < y < x, \\ 0 & \text{otherwise.} \end{cases}$$

Determine **(a)** the marginal joint p.d.f. of $Y_1, \ldots, Y_n$ and **(b)** the conditional p.d.f. of X for any given values of $Y_1, \ldots, Y_n$.

9. Let X be a random variable with a continuous distribution. Let $X_1 = X_2 = X$.

a. Prove that both X_1 and X_2 have a continuous distribution.

b. Prove that $\boldsymbol{X} = (X_1, X_2)$ does not have a continuous joint distribution.

10. Return to the situation described in Example 3.7.18. Let $\boldsymbol{X} = (X_1, \ldots, X_5)$ and compute the conditional p.d.f. of Z given $\boldsymbol{X} = \boldsymbol{x}$ directly in one step, as if all of $\boldsymbol{X}$ were observed at the same time.

11. Suppose that $X_1, \ldots, X_n$ are independent. Let $k < n$ and let $i_1, \ldots, i_k$ be distinct integers between 1 and n. Prove that $X_{i_1}, \ldots, X_{i_k}$ are independent.

12. Let $\boldsymbol{X}$ be a random vector that is split into three parts, $\boldsymbol{X} = (\boldsymbol{Y}, \boldsymbol{Z}, \boldsymbol{W})$. Suppose that $\boldsymbol{X}$ has a continuous joint distribution with p.d.f. $f(\boldsymbol{y}, \boldsymbol{z}, \boldsymbol{w})$. Let $g_1(\boldsymbol{y}, \boldsymbol{z}|\boldsymbol{w})$ be the conditional p.d.f. of $(\boldsymbol{Y}, \boldsymbol{Z})$ given $\boldsymbol{W} = \boldsymbol{w}$, and let $g_2(\boldsymbol{y}|\boldsymbol{w})$ be the conditional p.d.f. of $\boldsymbol{Y}$ given $\boldsymbol{W} = \boldsymbol{w}$. Prove that $g_2(\boldsymbol{y}|\boldsymbol{w}) = \int g_1(\boldsymbol{y}, \boldsymbol{z}|\boldsymbol{w})\, d\boldsymbol{z}$.

13. Let X_1, X_2, X_3 be conditionally independent given $Z = z$ for all z with the conditional p.d.f. $g(x|z)$ in Eq. (3.7.12). Also, let the marginal p.d.f. of Z be f_0 in Eq. (3.7.11). Prove that the conditional p.d.f. of X_3 given $(X_1, X_2) = (x_1, x_2)$ is $\int_0^\infty g(x_3|z)g_0(z|x_1, x_2)\, dz$, where g_0 is defined in Eq. (3.7.13). (You can prove this even if you cannot compute the integral in closed form.)

14. Consider the situation described in Example 3.7.14. Suppose that $X_1 = 5$ and $X_2 = 7$ are observed.

a. Compute the conditional p.d.f. of X_3 given $(X_1, X_2) = (5, 7)$. (You may use the result stated in Exercise 12.)

b. Find the conditional probability that $X_3 > 3$ given $(X_1, X_2) = (5, 7)$ and compare it to the value of $\Pr(X_3 > 3)$ found in Example 3.7.9. Can you suggest a reason why the conditional probability should be higher than the marginal probability?

15. Let $X_1, \ldots, X_n$ be independent random variables, and let W be a random variable such that $\Pr(W = c) = 1$ for some constant c. Prove that $X_1, \ldots, X_n$ are conditionally independent given $W = c$.

3.8 Functions of a Random Variable

Often we find that after we compute the distribution of a random variable X, we really want the distribution of some function of X. For example, if X is the rate at which customers are served in a queue, then $1/X$ is the average waiting time. If we have the distribution of X, we should be able to determine the distribution of $1/X$ or of any other function of X. How to do that is the subject of this section.